

Junghoon Suh

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EDUCATION

University of Southern California

Master of Science (Progressive Degree Program), Computer Science
Bachelor of Science, Computer Science / Business Administration

Los Angeles, CA, USA

Expected Graduation: December 2027

Graduation: May 2026

SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, C++, HTML/CSS, Swift, SQL

Technologies/Frameworks: Vue.js, React.js, React Native, Next.js, Spring Boot, PyTorch, Pandas, Flask, FastAPI, NumPy, Supabase, GitLab CI/CD, Jenkins, PostgreSQL, SQLite, DBeaver, scikit-learn, Redis, Kafka

EXPERIENCE

Exify+

April 2026 - August 2026

Founding Product/Software Engineering Intern

Los Angeles, CA (Remote)

- Replaced Exify's Excel valuation model with a full-stack simulator (**FastAPI, Next.js, TypeScript, SQLite**) enabling stakeholders to evaluate exit readiness across 2,000+ factor scores and 100 companies, validated with 95 automated tests
- Built an interactive decision-tree visualizer (**TypeScript, React Flow**) that traced each score change to its underlying rule and weight, making valuation adjustments auditable for non-technical stakeholders
- Owned requirements discovery across the CEO, 4 designers, marketing, and 6 SMB founders, translating user interviews and product discussions into scoring-engine and interface requirements

YD&S

September 2023 – August 2025

Software Engineer (Contract)

Seoul, South Korea

- Shipped and maintained features across 8 nationwide government data platforms each serving 1,000+ daily civil-service users; supported delivery through **GitLab and Jenkins** CI/CD pipelines
- Developed frontend interfaces for GRIP dashboards in **Vue.js/Vuex** and **JavaScript**, including an emergency alert system for the Environment Ministry, reducing rendering time by 60% (5s to 2s) through targeted optimization
- Designed backend modules for GRIP, including user-activity logging, data-aggregation pipelines, and dashboard services using **Java, Spring Boot, MyBatis, and SQL**, powering dashboards used by mayors and government agencies nationwide
- Led technical onboarding for 5 new engineers on Vue.js, Spring, and ETL, standardizing ramp-up on development workflows

RESEARCH

USC Intelligent and Cooperative Autonomy for Robotics (ICAROS) Lab

August 2026 - Present

Robotics Graduate Research Assistant

Los Angeles, CA (Remote)

- Investigating Vision-Language-Action models for language-conditioned manipulation in a 4-person team; building a feedback-driven system to close the VLM-to-action-policy grounding gap, targeting a top AI/robotics conference

USC Learning and Interactive Robot Autonomy (LIRA) Lab

September 2025 - May 2026

Undergraduate Research Intern

Los Angeles, CA

- Built a Python data pipeline for warm-start reinforcement learning, processing 2,500 offline demo transitions and supporting 100K-step pretraining followed by 50K-step online fine-tuning for robotic manipulation in JAX and ManiSkill
- Designed safety curricula with severity-tiered catastrophe regions and ran PPO experiments in PyTorch, NumPy, and ManiSkill across 12 configurations spanning 3 manipulation tasks to reduce catastrophe intrusions during policy rollout

PROJECTS

MatchPulse: Real-Time Agentic Soccer Intelligence Platform – ([Link](#))

[Python, FastAPI, Redpanda/Kafka, Redis, scikit-learn, Google ADK, Gemini API, Next.js/TypeScript]

- Engineered a distributed real-time football platform (FastAPI, Redpanda/Kafka, Redis) processing match events into tactical analysis via deterministic reducers; stress-tested to ~400 events/sec and 4.37s crash recovery
- Integrated 3 scikit-learn models over 380 Premier League matches, including a 136,930-possession retrieval index; built a seven-tool Gemini/ADK agent with a Next.js dashboard, achieving 30/30 grounding checks under adversarial prompts

Journi: Social Travel-Diary & Itinerary-Optimization Platform - ([Github](#))

[Spring Boot (Java), PostgreSQL, React Native, Python, C++]

- Built a social media/travel diary platform (Spring Boot, PostgreSQL, React Native) with a route-optimization engine (constrained TSP: greedy, 2-opt, simulated annealing) sequencing saved places into efficient itineraries
- Built a custom C++ key-value store with nonblocking TCP, incremental-rehashing hash tables, and min-heap TTL caching, reducing itinerary request latency $2.16\times$ (28ms $\rightarrow$ 13ms)